



University Of Engineering and Technology Peshawar
Jalozai campus

Computer programing

SUBMIDED BY

NAME	MUHAMMAD SULEMAN
ETEA ID	BF25NWELE0759
DEPARTMENT	ELECTRICAL ENGINEERING (POWER)
SECTION	A
Complex Engineering Probleum	1 ST
SEMESTER	2 nd
SESSION	FALL 2025

CEP SUBMITTED TO

Engr. Muhammad Rizwan sahib

COMPUTER PROGRAMMING Electrical Engineering Department — UET Nowshera

CEP LAB Lab Title:

Grid voltage monitoring system

Here are some simple and useful Power Grid Control programs in C++ for engineering practice.

1. Grid Voltage Monitoring System

This program checks voltage condition.

```
#include <iostream>

using namespace std;

int main()
{
    float voltage;

    cout << "Enter Grid Voltage: ";

    cin >> voltage;

    if(voltage > 240)
    {
        cout << "High Voltage Detected!" << endl;
    }

    else if(voltage < 210)
    {
        cout << "Low Voltage Detected!" << endl;
    }
}
```

```
    else
    {
        cout << "Voltage is Normal" << endl;
    }
    return 0;
}
```

2. Power Load Control System

```
#include <iostream>

using namespace std;

int main()
{
    int load;

    cout << "Enter Load in MW: ";

    cin >> load;

    if(load > 1000)
    {
        cout << "Grid Overloaded!" << endl;
        cout << "Start Backup Generator" << endl;
    }
    else
    {
        cout << "Grid Running Normally" << endl;
    }
}
```

```
    return 0;
}
```

3. Frequency Monitoring System

Pakistan grid frequency = 50 Hz

```
#include <iostream>
using namespace std;
int main()
{
    float frequency;
    cout << "Enter Frequency: ";
    cin >> frequency;

    if(frequency > 50)
    {
        cout << "Frequency is HIGH" << endl;
    }
    else if(frequency < 50)
    {
        cout << "Frequency is LOW" << endl;
    }
}
```

```
else
{
    cout << "Frequency is Stable" << endl;
}

return 0;
}
```

4. Transformer Protection Program

```
#include <iostream>
using namespace std;
int main()
{
    float temperature;
    cout << "Enter Transformer Temperature: ";
    cin >> temperature;

    if(temperature > 90)
    {
        cout << "Transformer Overheated!" << endl;
        cout << "Trip Breaker NOW!" << endl;
    }
    else
    {
        cout << "Transformer Safe" << endl;
    }
}
```

```
    return 0;
}
```

5. Automatic Generator Start System

```
#include <iostream>

using namespace std;

int main()
{
    bool powerFailure;

    cout << "Enter 1 for Power Failure or 0 for Normal: ";
    cin >> powerFailure;

    if(powerFailure)
    {
        cout << "Power Failed!" << endl;
        cout << "Generator Starting..." << endl;
    }
    else
    {
```

```
        cout << "Main Supply Running" << endl;
    }

    return 0;
}
```

6. Smart Street Light Control

```
#include <iostream>

using namespace std;

int main()
{
    int time;

    cout << "Enter Current Hour (0-23): ";

    cin >> time;

    if(time >= 18 || time <= 6)
    {
        cout << "Street Lights ON" << endl;
    }

    else
    {
        cout << "Street Lights OFF" << endl;
    }

    return 0;
}
```

}

Concepts Used

Concept

Used In

if-else

Decision making

Input/Output

User interaction

Variables

Store data

Boolean

True/False condition

Operators

Comparisons

Real Engineering Applications

These programs are similar to systems used in:

National Transmission and Despatch Company

Power stations

Grid stations

Industrial automation

Smart homes

Solar systems

SCADA systems

Complex Engineering Problem Kya Hota Hai?

Engineering ma jab:

Bohat zyada data ho

Real-time decisions leni ho

Safety important ho

Multiple machines control karni ho

Fault detect karna ho

Automatic response dena ho

to usay Complex Engineering Problem kehte hain.

Example:

Agar poore city ka power grid control karna ho:

Voltage stable rakhna

Load balance karna

Fault detect karna

Blackout rokna

Backup generator chalana

Yesab aik difficult engineering task hai.

Ab In Programs Ko Samjhty Hain

1. Voltage Monitoring Program
if(voltage > 240)

Ye Kya Kar Raha Hai?

Program voltage check karta hai.

Agar:

Voltage zyada ho → danger

Voltage kam ho → machines damage

Normal ho → system safe

Real Life Use

Grid station ma sensors continuously voltage monitor karte hain.

Agar voltage:

240V se zyada ho jaye

to breaker trip ho sakta hai.

Engineering Problem

Manual checking impossible hai because:

Thousands of lines hoti hain

Voltage har second change hota hai

Programming Kya Asan Karti Hai?

Computer automatically:

voltage check karta hai

warning deta hai

system protect karta hai

Human effort kam hota hai.

2. Load Control Program

```
if(load > 1000)
```

Meaning

Agar grid par load zyada ho gaya:

system overload ho sakta hai

transformer burn ho sakta hai

Program backup generator start karne ka decision deta hai.

Real Engineering Problem

Electricity demand har second change hoti hai:

Din ma kam

Raat ma zyada

Summer ma bohat high

Insan manually itni fast calculations nahi kar sakta.

C++ Ka Fayda

Computer:

load monitor karta hai

instantly decision leta hai

blackout prevent karta hai

3. Frequency Monitoring

Pakistan frequency:

?

Agar Frequency Change Ho

Motors damage ho sakte hain

Grid unstable ho sakta hai

Complete shutdown bhi ho sakta hai

Program Ka Kaam

Program continuously:

frequency measure karta hai

abnormal condition detect karta hai

Complex Problem Kyu Hai?

Poor country ka frequency balance karna:

multiple power plants

different cities

changing demand

Ye bohat difficult task hai.

3. Transformer Protection

if(temperature > 90)

Kya Ho Raha Hai?

Transformer overheat ho raha hai.

Program:

danger detect karta hai

breaker trip karta hai

Real Problem

Agar transformer burn ho:

crore ka damage

area blackout

fire risk

Automation Ka Fayda

Computer:

24/7 monitoring karta hai

milliseconds ma response deta hai

Human itni speed se react nahi kar sakta.

5. Generator Start System

Problem

Agar power fail ho jaye:

hospitals band

factories stop

servers shutdown

Program Solution

Program:

power failure detect karta hai

generator automatically start karta hai

Real Engineering Systems

Ye logic use hota hai:

Hospitals

Airports

Data centers

Grid stations

Industries

Programming Engineering Ko Kaisa Easy Banati Hai?

Without Programming

With Programming

Manual monitoring

Automatic monitoring

Slow response

Instant response

Human errors

Accurate decisions

More workers needed

Less manpower

Difficult calculations

Fast calculations

Real Technologies

Real systems use:

C++

PLC

SCADA

Embedded systems

AI

Companies:

Siemens

ABB

Schneider Electric

Simple Understanding

Ye programs asal ma:

Data lete hain

Condition check karte hain

Decision lete hain

Action perform karte hain